RACE TO THE BOTTOM: COMPETITION AND QUALITY IN SCIENCE*

RYAN HILL AND CAROLYN STEIN

This article investigates how competition to publish first and thereby establish priority affects the quality of scientific research. We begin by developing a model where scientists decide whether and how long to work on a given project. When deciding how long they should let their projects mature, scientists trade off the marginal benefit of higher-quality research against the marginal risk of being preempted. Projects with the highest scientific potential are the most competitive because they induce the most entry. Therefore, the model predicts these projects are also the most rushed and lowest quality. We test the predictions of this model in the field of structural biology using data from the Protein Data Bank (PDB), a repository for structures of large macromolecules. An important feature of the PDB is that it assigns objective measures of scientific quality to each structure.